

Regime-Switching Asset Allocation via a POMDP

Type: New Application

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The problem: regimes break static allocation

The 60/40 portfolio lost $\sim 17\%$ in 2022.

A static allocation assumes stable joint return moments. Markets violate this in two ways:

- **Conditional heteroskedasticity:** long quiet periods broken by short crises.
- **Stress correlation:** stocks and bonds sell off together when it hurts most.

Question. Can a decision-theoretic regime overlay, a POMDP solved end-to-end on macroeconomic observables, beat a static benchmark after realistic frictions?

Why POMDP, not a tree, ID, or MDP

Naive choice	Why it fails here
Decision tree	Tree explodes with horizon; no shared state.
Influence diagram	Same blow-up; not natural for repeated rebalance.
Fully-observable MDP	Requires knowing the regime, but it is <i>latent</i> .
POMDP	Hidden state, Bayesian belief, QMDP for tractable control.

Tools assembled: Bayesian forward filter; Markov chain over regimes; infinite-horizon Bellman recursion solved by value iteration and exact policy iteration; structural results for monotone policies.

POMDP tuple, written once and reused

$\mathcal{T} = \{0, 1, 2, \dots\}$, monthly epochs.

$\mathcal{S} = \{s_1, \dots, s_K\}$ latent regimes ($K = 2, 3, 4$).

$\mathcal{A} = \{(0/100), (20/80), \dots, (100/0)\}$ discrete portfolio grid.

$T(s' | s)$ HMM transition matrix; **action-independent** (regimes are exogenous).

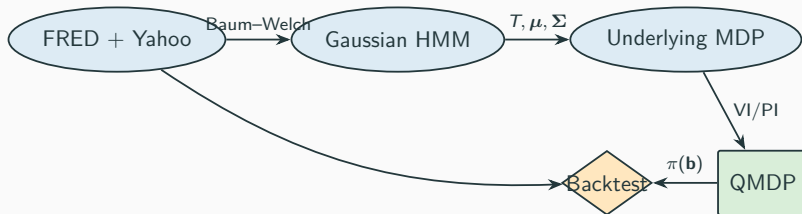
$O(\mathbf{o} | s)$ Gaussian emission on $\mathbf{o} = (\text{VIX}, \text{term spread})$.

$R(s, \mathbf{a}) = \mathbb{E}[U(\mathbf{a}^\top \mathbf{r}) | s] - c\|\mathbf{a} - \mathbf{a}_{\text{prev}}\|_1$, CRRA utility.

$\lambda = 0.95$ monthly discount.

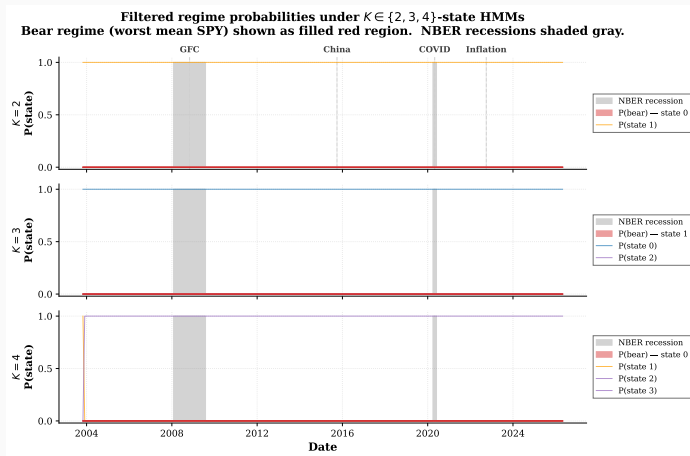
Belief update (Bayesian filter): $\mathbf{b}_t(s') \propto O(\mathbf{o}_t | s') \sum_s T(s' | s) \mathbf{b}_{t-1}(s)$.

Pipeline



- Raw observations $\rightarrow$ HMM emissions and transitions.
- HMM transition matrix becomes the MDP's T . Reward built by MC.
- VI and PI cross-checked. QMDP folds belief into Q^* .
- Backtest is a walk-forward filter+act loop on real history.

HMM regime timeline



Worst-mean-SPY regime in red. NBER bars in gray. The 2-state model recovers 2008, 2015–16, 2020–23 stress.

Underlying MDP: VI and PI agree

The fully-observable MDP satisfies

$$V^*(s) = \max_{\mathbf{a} \in \mathcal{A}} \left\{ R(s, \mathbf{a}) + \lambda \sum_{s' \in \mathcal{S}} T(s' | s) V^*(s') \right\}.$$

Value iteration:

$V^{(n+1)} = L V^{(n)}$, stop at

$$\|V^{(n+1)} - V^{(n)}\|_{\infty} < \frac{\varepsilon(1-\lambda)}{2\lambda}.$$

125 iters for $\lambda = 0.95$, $\varepsilon = 10^{-4}$.

Policy iteration:

solve $(I - \lambda P_{\pi})V = r_{\pi}$, then
greedy improvement.

2 iters to converge.

Proposition. VI greedy policy = PI greedy policy, values agree within 2.6×10^{-6} .

QMDP: project belief onto Q^*

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{s \in \mathcal{S}} \mathbf{b}(s) Q^*(s, \mathbf{a}).$$

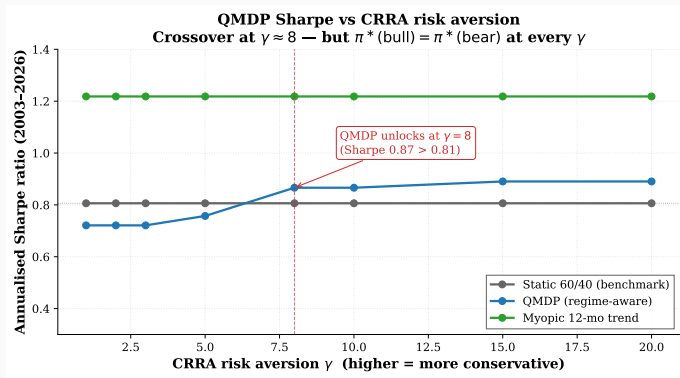
- Exact in the fully-observable limit ($\mathbf{b}$ concentrated).
- Treats POMDP as belief-weighted average of MDPs.
- Avoids the intractable belief-space Bellman recursion.
- Implementation: `src/models/qmdp.py::qmdp_action`, single $O(K)$ inner product.

Backtest 2003–2026, headline

Policy	CAGR	Vol	Sharpe	Max DD	Calmar
Static 60/40	7.35%	9.4%	0.81	−34.2%	0.22
QMDP ($\gamma = 2$)	9.90%	14.7%	0.72	−53.0%	0.19
Myopic 12-mo	13.99%	11.3%	1.22	−21.1%	0.66

Finding. QMDP under log utility *collapses to 100% stocks*. Risk-tolerance, not regime detection, drives the result. The myopic 12-month trend follower wins on risk-adjusted terms.

Sensitivity: QMDP unlocks at $\gamma \geq 8$



However: π^* (bull) = π^* (bear) at every γ . The gains are risk-tolerance shifts, not regime tilts.

Conclusions & extensions

What we showed.

- Pipeline correct: VI+PI agree, HMM recovers 2008/2020/2022 stress.
- Under CRRA $\gamma \geq 8$, QMDP Sharpe (0.87–0.89) beats static (0.81).
- But the regime *signal* is detectable while the regime *action* is not, given two macro observables.

Extensions (in priority order).

1. **Richer observations:** restore HY OAS via FRED API key.
2. **Sharpe / CVaR reward shaping:** explicitly penalise bear-regime variance.
3. **Point-based VI:** exploit information value on a 1-D belief simplex (cheap).

Code, data and pipeline: github.com/takakhoo/ENGS177_Final_Project